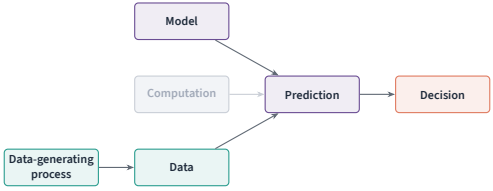
ACCELERATE PROGRAMME FOR SCIENTIFIC DISCOVERY

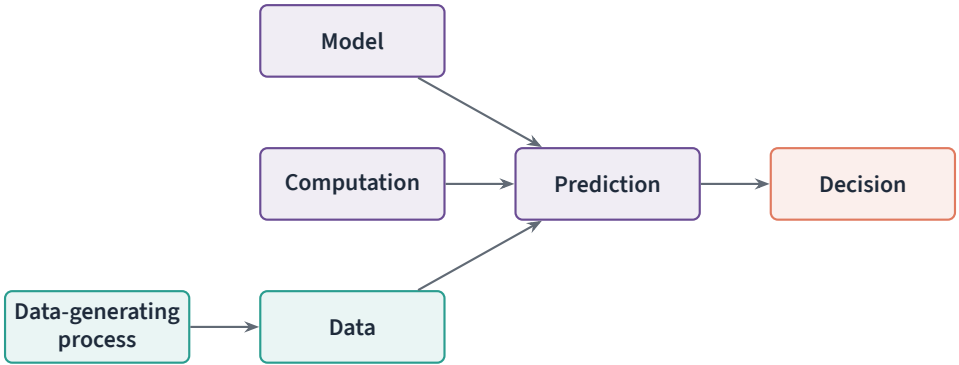
Uncertainty in AI

So far

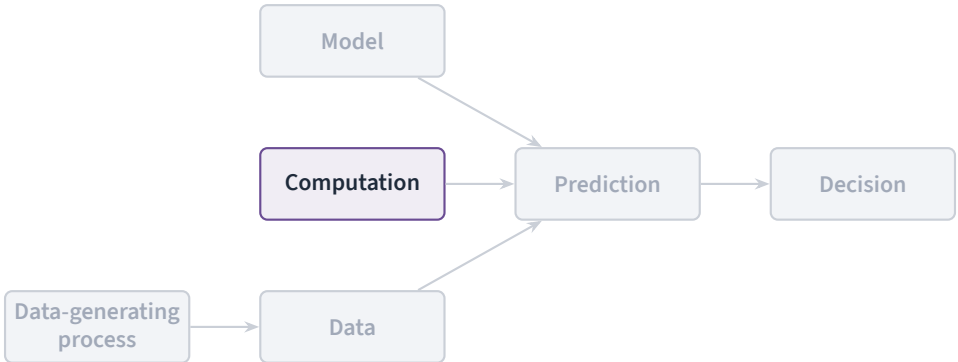


data-generating process	may be random: $p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$
data	samples from it: what, how and how much we measure
model	$p(y_\star, \mathcal{D}, \theta \mid x_\star) = p(y_\star \mid x_\star, \theta)\, p(\mathcal{D} \mid \theta)\, p(\theta)$
prediction	$p(y_\star \mid x_\star, \mathcal{D}) = \int p(y_\star \mid x_\star, \theta)\, p(\theta \mid \mathcal{D})\, d\theta$
decision	$\ell(a, y_\star)$: the cost of each action in each scenario
	$a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$
uncertainty	$\int \ell(a_{\text{opt}}, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$

Next: computation

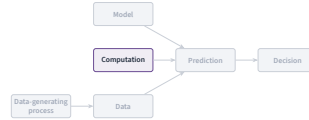


Next: computation



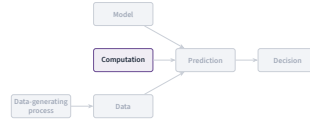
Shortcuts we take to get answers in reasonable time

Prediction: model, data and computation



$$p(y_\star \mid x_\star, \mathcal{D}) = \int \underbrace{p(y_\star \mid x_\star, \theta)}_{\text{model}} \underbrace{p(\theta \mid \mathcal{D})}_{\text{model and data}} \, d\theta$$

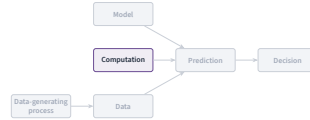
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computation

Prediction: model, data and computation

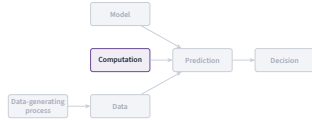


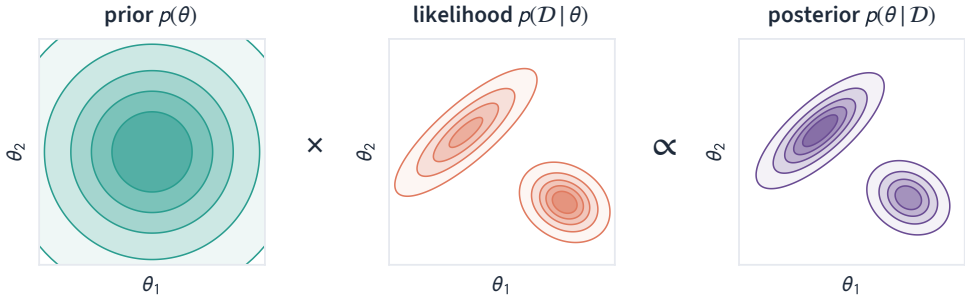
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computation

The methods differ in how they approximate the posterior and the integral

The posterior, as a landscape

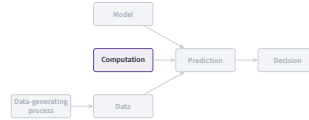


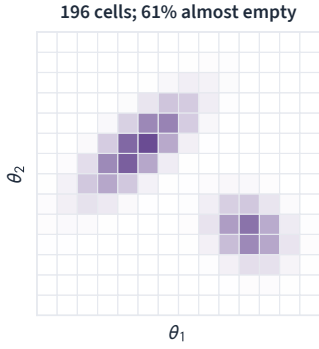


a cartoon: two of the network's 82 numbers, θ_1 and θ_2

High posterior: settings that explain the data and were plausible beforehand

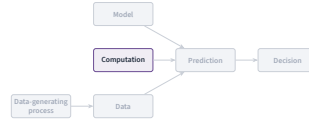
Evaluating everywhere: a grid

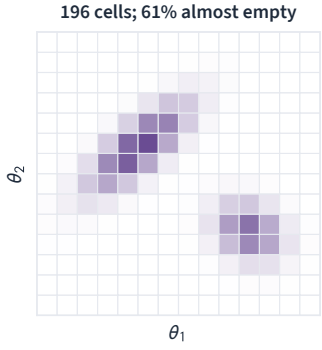




the direct approach: a sum over a grid of settings, each weighted by its posterior

Evaluating everywhere: a grid

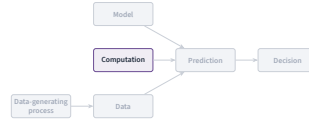


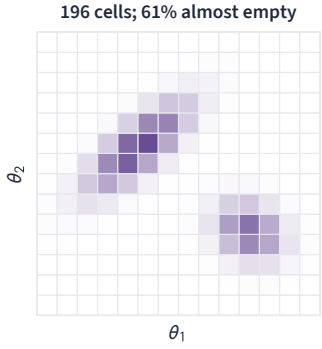


the direct approach: a sum over a grid of settings, each weighted by its posterior

our network has 82 numbers: 10 values each gives 10^{82} cells

Evaluating everywhere: a grid



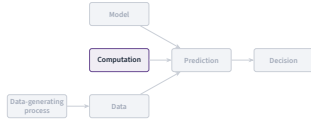


the direct approach: a sum over a grid of settings, each weighted by its posterior

our network has 82 numbers: 10 values each gives 10^{82} cells

Too many settings: we must choose a few good ones

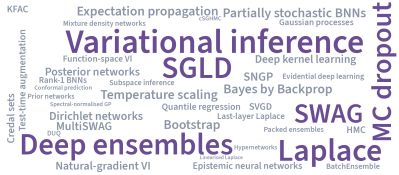
Two approximations, one recipe



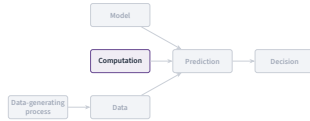
Two approximations

$$p(y_\star \mid x_\star, \mathcal{D}) = \int p(y_\star \mid x_\star, \theta)\; p(\theta \mid \mathcal{D})\, d\theta$$

each method approximates the integral and the posterior



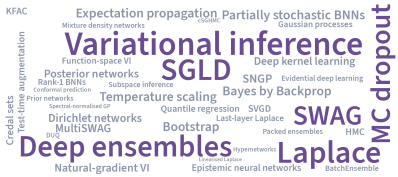
Two approximations, one recipe



Two approximations

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each method approximates the integral and the posterior

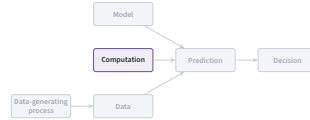


One recipe

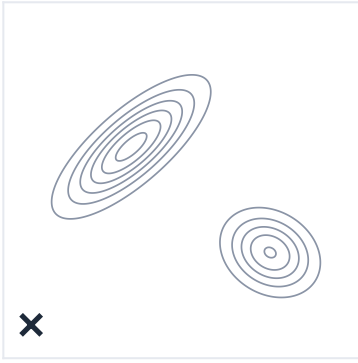
choose settings $\theta^{(1)}, \ldots, \theta^{(S)}$, and average their predictions

$$p(y_\star \mid x_\star, \mathcal{D}) \approx \frac{1}{S} \sum_{s=1}^{S} p(y_\star \mid x_\star, \theta^{(s)})$$

The gradient as a guide



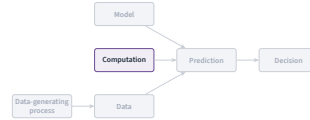
a starting setting



How do we find plausible settings?

$$\log p(\theta \mid \mathcal{D}) = \log p(\mathcal{D} \mid \theta) + \log p(\theta) - \log p(\mathcal{D})$$

The gradient as a guide



the gradient: steepest way up



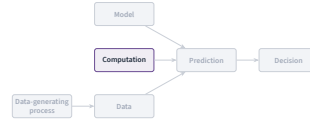
How do we find plausible settings?

$$\log p(\theta \mid \mathcal{D}) = \log p(\mathcal{D} \mid \theta) + \log p(\theta) - \log p(\mathcal{D})$$

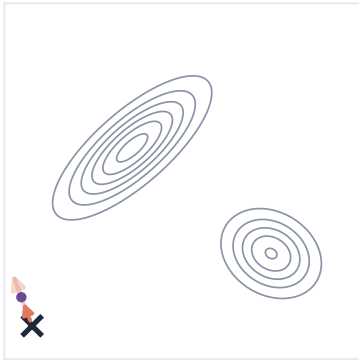
The gradient points to more plausible settings

$$\nabla_\theta \log p(\theta \mid \mathcal{D}) = \nabla_\theta \log p(\mathcal{D} \mid \theta) + \nabla_\theta \log p(\theta)$$

The gradient as a guide



one small step along it



How do we find plausible settings?

$$\log p(\theta \mid \mathcal{D}) = \log p(\mathcal{D} \mid \theta) + \log p(\theta) - \log p(\mathcal{D})$$

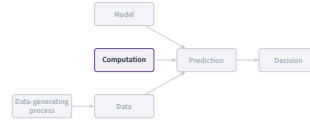
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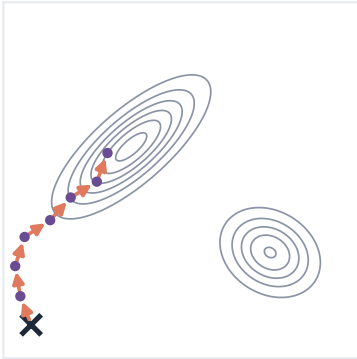
Take a step

$$\theta \leftarrow \theta + \epsilon \nabla_\theta \log p(\theta \mid \mathcal{D})$$

The gradient as a guide



repeat: small steps up to a peak



How do we find plausible settings?

$$\log p(\theta \mid \mathcal{D}) = \log p(\mathcal{D} \mid \theta) + \log p(\theta) - \log p(\mathcal{D})$$

The gradient points to more plausible settings

$$\nabla_\theta \log p(\theta \mid \mathcal{D}) = \nabla_\theta \log p(\mathcal{D} \mid \theta) + \nabla_\theta \log p(\theta)$$

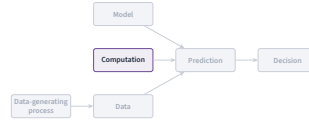
Take a step

$$\theta \leftarrow \theta + \epsilon \nabla_\theta \log p(\theta \mid \mathcal{D})$$

Keep climbing: the peak

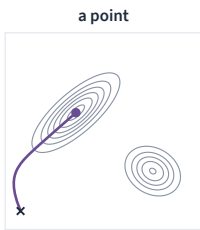
$$\theta_{\text{MAP}} = \arg\max_\theta \; p(\theta \mid \mathcal{D})$$

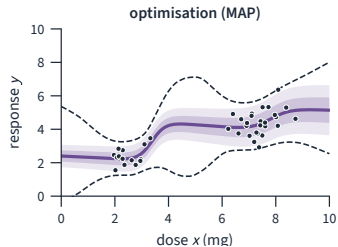
Optimisation: a point, MAP



Follow the gradient to the most plausible setting: the best choice if we can keep only one

$$\theta \leftarrow \theta + \epsilon \nabla_\theta \log p(\theta \mid \mathcal{D}) \qquad p(y_\star \mid x_\star, \mathcal{D}) \approx p(y_\star \mid x_\star, \theta_{\text{MAP}})$$

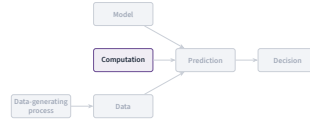




left: a cartoon with 2 settings instead of 82 ($\times$: start) right, dashed: an impractical but accurate reference (backup)

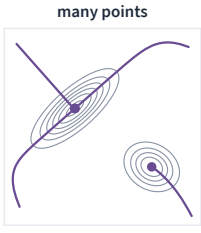
In short: the average is exact integration, with posterior $\approx$ a point at a peak

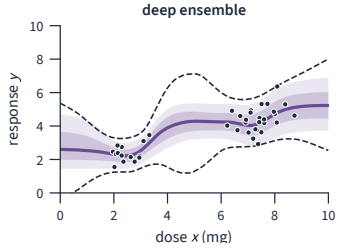
Optimisation: many points, deep ensembles



Repeat the optimisation from M different starts, and average the predictions

$$\theta^{(m)} \leftarrow \theta^{(m)} + \epsilon \nabla_\theta \log p(\theta^{(m)} \mid \mathcal{D}) \qquad p(y_\star \mid x_\star, \mathcal{D}) \approx \tfrac{1}{M} \sum_{m=1}^{M} p(y_\star \mid x_\star, \theta^{(m)})$$

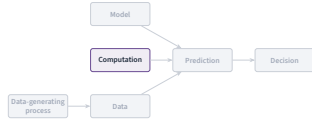




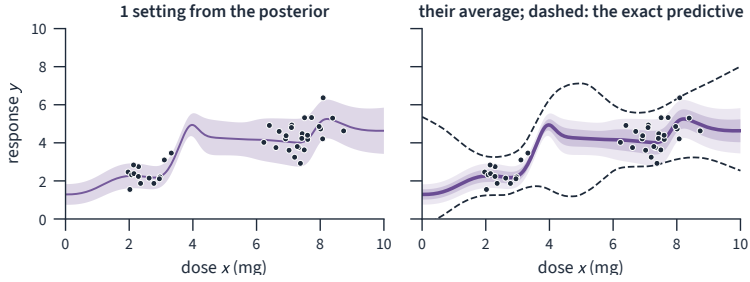
In short: the average is exact integration, with posterior ≈ a few points at peaks

B. Lakshminarayanan, A. Pritzel and C. Blundell (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. *NeurIPS*.

Sampling: Monte Carlo



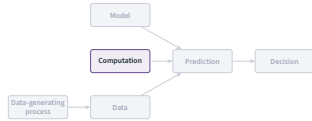
$$p(y_\star \mid x_\star, \mathcal{D}) \;\approx\; \frac{1}{S}\sum_{s=1}^{S} p(y_\star \mid x_\star, \theta^{(s)}), \qquad \theta^{(s)} \text{ drawn from } p(\theta \mid \mathcal{D})$$



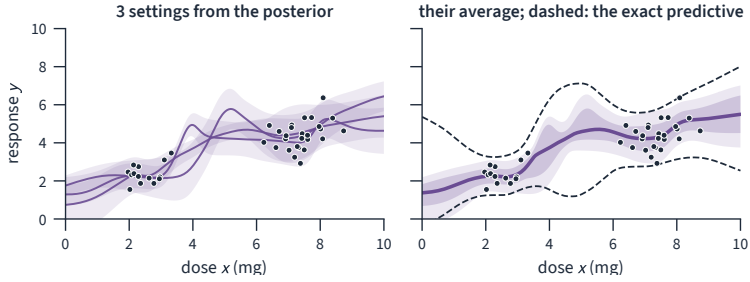
More samples: the average approaches the exact predictive

The samples here come from HMC: R. M. Neal (2011). MCMC using Hamiltonian dynamics. *Handbook of Markov Chain Monte Carlo*.

Sampling: Monte Carlo



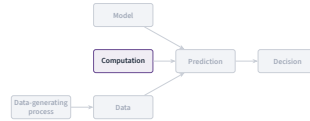
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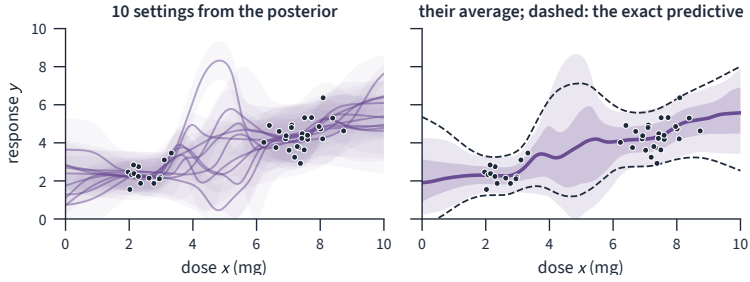
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Sampling: Monte Carlo



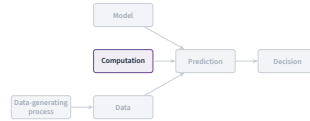
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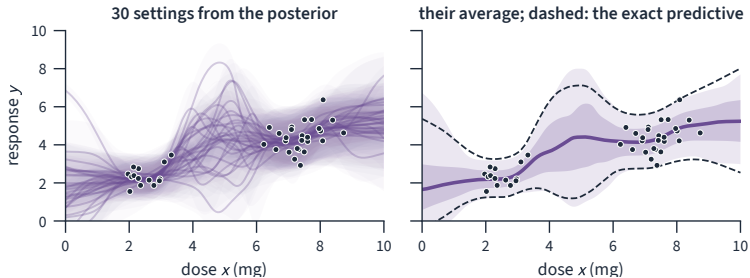
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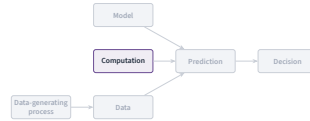
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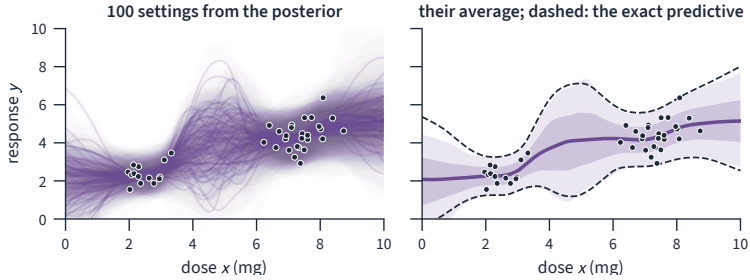
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Sampling: Monte Carlo



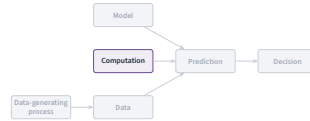
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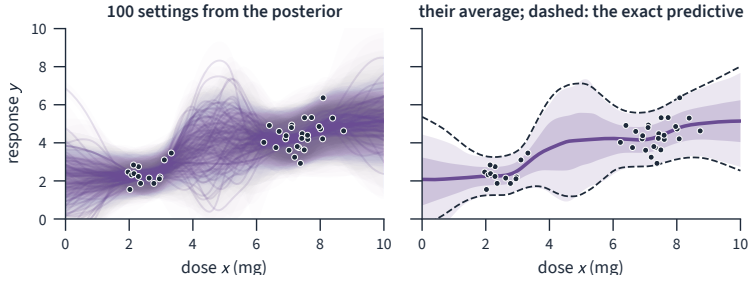
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The samples here come from HMC: R. M. Neal (2011). MCMC using Hamiltonian dynamics. *Handbook of Markov Chain Monte Carlo*.

Sampling: Monte Carlo



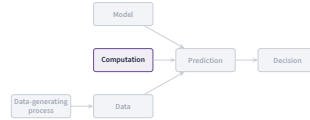
$$p(y_\star \mid x_\star, \mathcal{D}) \;\approx\; \frac{1}{S}\sum_{s=1}^{S} p(y_\star \mid x_\star, \theta^{(s)}), \qquad \theta^{(s)} \text{ drawn from } p(\theta \mid \mathcal{D})$$



In practice: approximate samples, from a stand-in (Laplace, VI) or a cheaper process (SGLD)

The samples here come from HMC: R. M. Neal (2011). MCMC using Hamiltonian dynamics. *Handbook of Markov Chain Monte Carlo*.

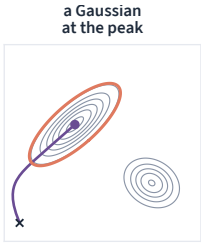
Optimise, then sample: a Gaussian at the peak, Laplace

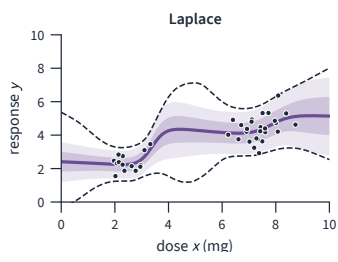


From settings to a distribution: a Gaussian around θ_{MAP}

$$p(\theta \mid \mathcal{D}) \approx \mathcal{N}\left(\theta_{\text{MAP}}, H^{-1}\right), \quad H = -\nabla^2_\theta \log p(\theta \mid \mathcal{D})\big|_{\theta_{\text{MAP}}}$$

$$p(y_\star \mid x_\star, \mathcal{D}) \approx \frac{1}{S}\sum_s p(y_\star \mid x_\star, \theta^{(s)})$$

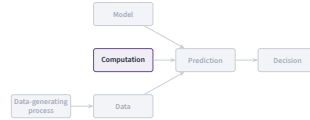




In short: the average is approximate Monte Carlo integration, with posterior ≈ a Gaussian at a peak

D. J. C. MacKay (1992), *Neural Computation*; E. Daxberger et al. (2021), Laplace redux, *NeurIPS*.

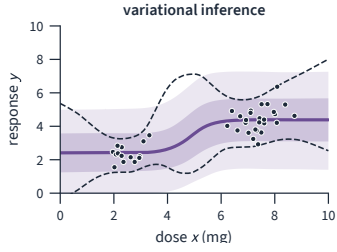
Optimise, then sample: a moving distribution, VI



Fit a simple distribution q_ϕ to the posterior by following the gradient of their distance

$$\phi \leftarrow \phi - \epsilon \nabla_\phi \mathrm{KL}\big(q_\phi \,\big\|\, p(\theta \mid \mathcal{D})\big) \qquad p(y_\star \mid x_\star, \mathcal{D}) \approx \tfrac{1}{S} \textstyle\sum_s p(y_\star \mid x_\star, \theta^{(s)})$$

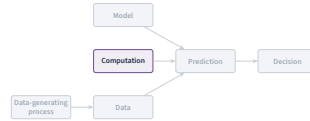




In short: the average is approximate Monte Carlo integration, with posterior ≈ the closest Gaussian

C. Blundell, J. Cornebise, K. Kavukcuoglu and D. Wierstra (2015). Weight uncertainty in neural networks. *ICML*.

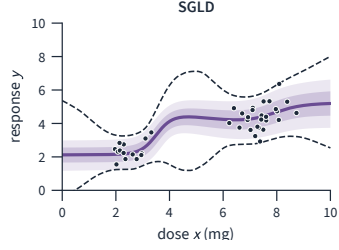
Sampling: a noisy point, SGLD



Instead of a Gaussian, let the settings wander through the posterior itself: with noise in the gradient steps, the path visits settings in proportion to their plausibility

$$\theta \leftarrow \theta + \tfrac{\epsilon}{2}\,\widehat{\nabla}_\theta \log p(\theta \mid \mathcal{D}) + \eta, \quad \eta \sim \mathcal{N}(0, \epsilon I) \qquad p(y_\star \mid x_\star, \mathcal{D}) \approx \tfrac{1}{S}\sum_s p(y_\star \mid x_\star, \theta^{(s)})$$

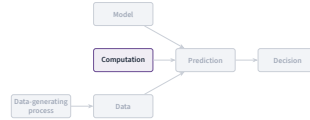




In short: the average is approximate Monte Carlo integration, with posterior ≈ (almost) itself

M. Welling and Y. W. Teh (2011). Bayesian learning via stochastic gradient Langevin dynamics. *ICML*.

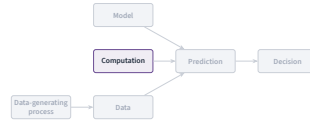
Five ways, compared



Every method: choose some settings, and average their predictions

method	chosen settings
optimisation	
MAP	one peak
deep ensembles	several peaks
optimise, then sample	
Laplace	draws from a Gaussian at a peak
VI	draws from a fitted Gaussian
sampling	
SGLD	points along a noisy path

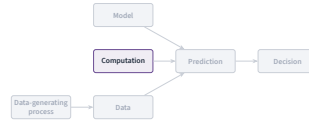
Five ways, compared



Every method: choose some settings, and average their predictions

method	chosen settings	posterior ≈
optimisation		
MAP	one peak	a point
deep ensembles	several peaks	a few points
optimise, then sample		
Laplace	draws from a Gaussian at a peak	a Gaussian at a peak
VI	draws from a fitted Gaussian	the closest Gaussian
sampling		
SGLD	points along a noisy path	(almost) itself

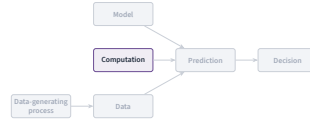
Five ways, compared

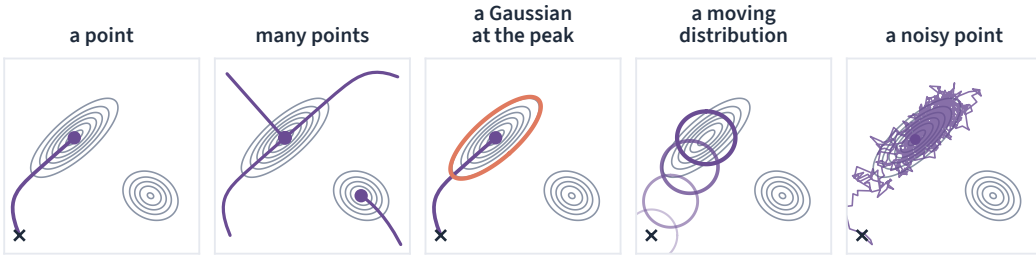


Every method: choose some settings, and average their predictions

method	chosen settings	posterior ≈	integral approx.
optimisation			
MAP	one peak	a point	none
deep ensembles	several peaks	a few points	none
optimise, then sample			
Laplace	draws from a Gaussian at a peak	a Gaussian at a peak	Monte Carlo
VI	draws from a fitted Gaussian	the closest Gaussian	Monte Carlo
sampling			
SGLD	points along a noisy path	(almost) itself	Monte Carlo

Five ways, one landscape

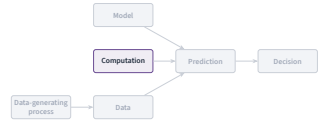


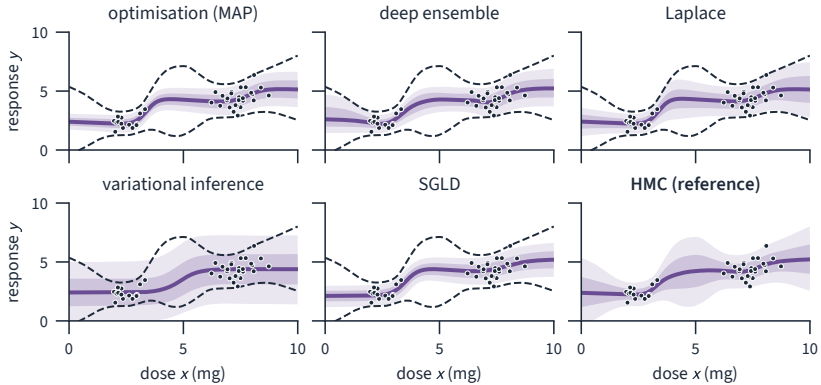


a cartoon: two settings instead of our network's 82; contours show the posterior; × marks where training starts

Each method finds or describes plausible settings in its own way

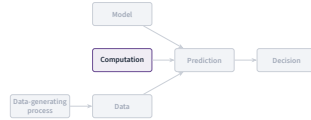
Comparing the predictions

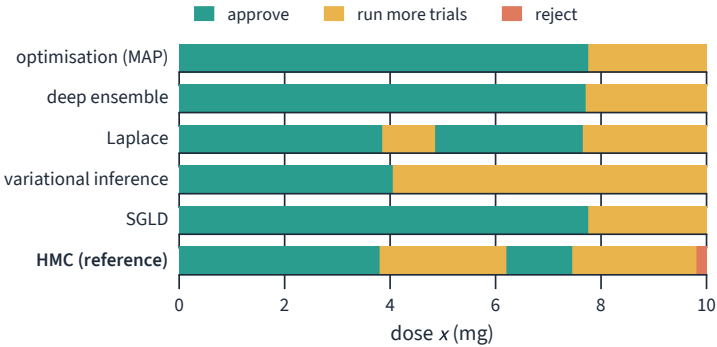




Same model, same data: different approximations, different predictions

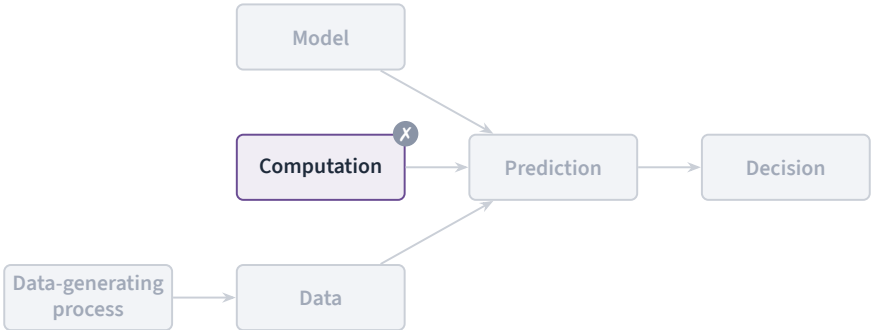
Does the method change the decision?





Same model, same data: the approximation changes the decision

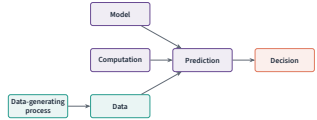
What computation doesn’t capture



Approximation error is not part of the prediction

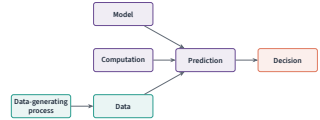
each method reports its own uncertainty, not how far it is from the exact answer

Recap



data-generating process $p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$

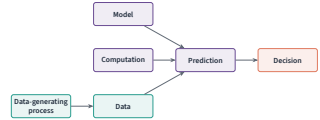
Recap



data-generating process $p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$

data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, samples from the true process

Recap

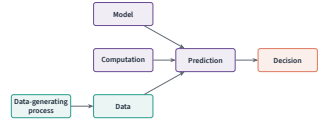


data-generating process $p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$

data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, samples from the true process

model $p(y_\star, \mathcal{D}, \theta \mid x_\star) = p(y_\star \mid x_\star, \theta)\, p(\mathcal{D} \mid \theta)\, p(\theta)$

Recap



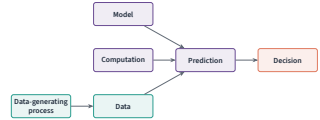
data-generating process $p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$

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model $p(y_\star, \mathcal{D}, \theta \mid x_\star) = p(y_\star \mid x_\star, \theta)\, p(\mathcal{D} \mid \theta)\, p(\theta)$

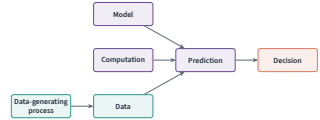
prediction $p(y_\star \mid x_\star, \mathcal{D}) = \int p(y_\star \mid x_\star, \theta)\, p(\theta \mid \mathcal{D})\, d\theta$

Recap



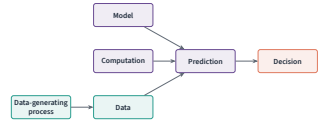
data-generating process	$p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$
data	$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, samples from the true process
model	$p(y_\star, \mathcal{D}, \theta \mid x_\star) = p(y_\star \mid x_\star, \theta)\, p(\mathcal{D} \mid \theta)\, p(\theta)$
prediction	$p(y_\star \mid x_\star, \mathcal{D}) = \int p(y_\star \mid x_\star, \theta)\, p(\theta \mid \mathcal{D})\, d\theta$
computation	$\approx \frac{1}{S} \sum_s p(y_\star \mid x_\star, \theta^{(s)})$, with $\theta^{(s)}$ from an approximation

Recap



data-generating process	$p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$
data	$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, samples from the true process
model	$p(y_\star, \mathcal{D}, \theta \mid x_\star) = p(y_\star \mid x_\star, \theta)\, p(\mathcal{D} \mid \theta)\, p(\theta)$
prediction	$p(y_\star \mid x_\star, \mathcal{D}) = \int p(y_\star \mid x_\star, \theta)\, p(\theta \mid \mathcal{D})\, d\theta$
computation	$\approx \frac{1}{S} \sum_s p(y_\star \mid x_\star, \theta^{(s)})$, with $\theta^{(s)}$ from an approximation
decision	$\ell(a, y_\star)$

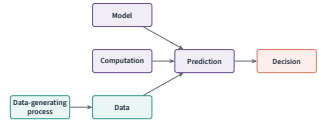
Recap

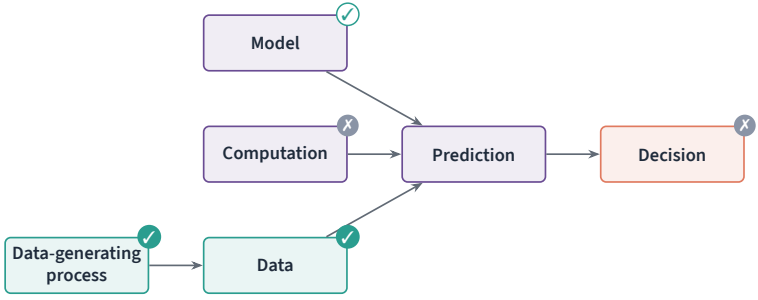


data-generating process	$p_{\text{true}}(x, y) = p_{\text{true}}(y \mid x)\, p_{\text{true}}(x)$
data	$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, samples from the true process
model	$p(y_\star, \mathcal{D}, \theta \mid x_\star) = p(y_\star \mid x_\star, \theta)\, p(\mathcal{D} \mid \theta)\, p(\theta)$
prediction	$p(y_\star \mid x_\star, \mathcal{D}) = \int p(y_\star \mid x_\star, \theta)\, p(\theta \mid \mathcal{D})\, d\theta$
computation	$\approx \frac{1}{S} \sum_s p(y_\star \mid x_\star, \theta^{(s)})$, with $\theta^{(s)}$ from an approximation
decision	$\ell(a, y_\star)$
	$a_{\text{opt}} = \arg\min_a \int \ell(a, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$
uncertainty	$\int \ell(a_{\text{opt}}, y_\star)\, p(y_\star \mid x_\star, \mathcal{D})\, dy_\star$

Everything hinges on the prediction $p(y_\star \mid x_\star, \mathcal{D})$

What this workshop captured



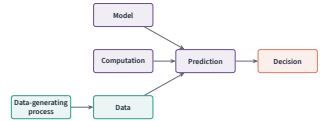


Randomness in the process and limited data, as uncertainty about the model's settings

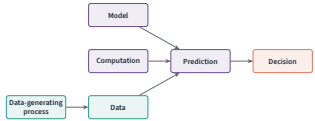
not captured: what we chose not to model, the model class, approximation error, the loss itself

✓ captured ✓ partly ✗ not captured

Not covered today



- **misspecification**: when no setting of the model describes the true process
- **missing data**: when some doses or responses are not recorded
- **model selection**: choosing or averaging over model classes M
- **conformal prediction**: prediction sets with coverage guarantees, without a prior
- **temperature scaling**: recalibrating a trained network's probabilities
- **evidential deep learning**: predicting the parameters of a distribution over distributions
- **distance-aware methods**: Gaussian processes and SNGP, which notice unusual inputs
- **generalised Bayes**: replacing the likelihood with a loss



Tomorrow: the practical

Questions?